WINTER 2025

SOIL HEALTH & CROP NUTRITION PLAN

FH Moses & Sons Pty Ltd

Prepared By : Sam Gulliford

TRADING NAME: FH Moses & Sons Pty Ltd
ACCOUNT NUMBER:
FARM NAME: Valais
CONTACT: James Moses
AREA (ha): not provided
DATE: 27 May 2025

ACCREDITED ADVISER: Sam Gulliford
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LABORATORY: Nutrient Advantage Lab

Paddock Name	**Lane**	**Lane**
Target production (t/ha)	**3.0**	**0.0**
Laboratory sample number	79677915	79671236
Profile sampled (cm)	0-10	10-90
Date of sampling	26 Mar 2025	26 Mar 2025
Evaluation table	Canola, Northern Australia, 2 - 4 t/ha, PBI (70-280)	Canola, Northern Australia, 2 - 4 t/ha, PBI (70-280)
Description	Lane	Lane

ANALYSIS RESULTS

Paddock Name	Lane	Lane
Sample Depth (cm)	**0-10**	**10-90**
Soil texture	Clay Loam	Clay Loam
Soil colour	Red Brown	
Soil Colour (Munsell)	26	
Colour (SM Code)	26	
Soil Texture (Hand bolus)	5	
pH (1:5 H2O)	6.3 - Satisfactory	7.9 - High
pH (1:5 CaCl2)	6 - Satisfactory	7.2 - High
EC (1:5 H2O) dS/m	0.22 - Satisfactory	0.07 - Satisfactory
EC (se) (dS/m)	1.8 - Satisfactory	0.6 - Satisfactory
EC (se) (dS/m) (Cladj)	0.8 - Satisfactory	0.3 - Satisfactory
Chloride (1:5 H2O) mg/kg	10 - Satisfactory	10 - Satisfactory
Organic carbon (Walkley Black) %	3.79 - High	
Nitrate-N (Group) mg/kg	46 - Sufficient	3 - Low
Ammonium nitrogen (KCl) mg/kg	6 - High	2 - Sufficient
Phosphorus (Colwell) mg/kg	170 - High	
Phosphorus (BSES) mg/kg	190 - Sufficient	
Phosphorus Buffer Index (Colwell) (PBIc)	300 - Sufficient	
Phosphorus Environmental Risk Index	0.6 - Satisfactory	
Potassium (Amm-Acet.) cmol+/kg	1.9 - High	
Potassium % of CEC	7.5 - Sufficient	
Sulfate-S (KCl40) mg/kg	9.2 - Sufficient	2.1 - Low
Calcium (Amm-Acet) cmol+/kg	18 - Sufficient	
Calcium % of CEC	70.8 - Sufficient	
Magnesium (Amm-Acet.) cmol+/kg	5.5 - Sufficient	
Magnesium % cations	21.6 - Satisfactory	-
Calcium:magnesium ratio	3.3	
Calcium:Magnesium Ratio (cmol+/kg)	3.3	
Sodium (Amm-Acet.) cmol+/kg	0.03 - Sufficient	
Sodium % cations		-
Exch. sodium %	0.1 - Satisfactory	
Electrochemical Stability Index	1.805 - Satisfactory	-
Aluminium (KCl) cmol+/kg	0.1 - Sufficient	
Aluminium Saturation %	1 - Sufficient	
eCEC cmol+/kg	25.4 - Satisfactory	-
Copper (DTPA) mg/kg	3 - Sufficient	
Zinc (DTPA) mg/kg	1.4 - Sufficient	
Manganese (DTPA) mg/kg	110 - Excess	
Iron (DTPA) mg/kg	90	
Boron (hot CaCl2) (mg/kg)	0.7 - Sufficient	

NUTRIENT REQUIREMENTS

Paddock Name	Lane	Lane
Area represented (ha)	not provided	not provided
Target production (t/ha)	**3**	**0**
Growing Season	2025 Winter	2025 Winter
NITROGEN (kg/ha)	**(116.6)**	
PHOSPHORUS (kg/ha)	**(13.1)**	

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- Lane

PINNACLE AGRICULTURE

SULFUR (kg/ha)	**(16.2)**	
ZINC (kg/ha)	**(0.60)**	

SOIL HEALTH		**Result**	**Low**	**Marginal**	**Sufficient**	**High**	**Excess**	**Sufficiency Range**
Soil Colour (Munsell)	0 - 10	26.0						
pH (1:5 H2O)	0 - 10	6.3						5.5 - 7.5
	10 - 90	7.9						5.5 - 7.5
pH (1:5 CaCl2)	0 - 10	6.0						4.7 - 6.7
	10 - 90	7.2						4.7 - 6.7
EC (1:5 H2O) dS/m	0 - 10	0.22						0.00 - 0.60
	10 - 90	0.07						0.00 - 0.60
EC (se) (dS/m)	0 - 10	1.8						0.0 - 9.7
	10 - 90	0.6						0.0 - 9.7
EC (se) (dS/m) (Cladj)	0 - 10	0.8						0.0 - 9.7
	10 - 90	0.3						0.0 - 9.7
Chloride (1:5 H2O) mg/kg	0 - 10	10						0 - 250
	10 - 90	10						0 - 250
Organic carbon (Walkley Black) %	0 - 10	3.79						1.00 - 2.00
Phosphorus Environmental Risk Index	0 - 10	0.60						0.00 - 0.65
Sodium:Potassium Ratio	0 - 10	0.0						0.0 - 5.0
Magnesium % cations	0 - 10	21.6						0.0 - 25.0
	10 - 90							
Calcium:magnesium ratio	0 - 10	3.3						
Calcium:Magnesium Ratio (cmol+/kg)	0 - 10	3.3						
Sodium % cations	10 - 90							
Exch. sodium %	0 - 10	0.1						0.0 - 6.0
Electrochemical Stability Index	0 - 10	1.805						0.050 - 10.000
	10 - 90							
Dispersion Index (Loveday/Pyle)	0 - 10	0						0 - 6
eCEC cmol+/kg	0 - 10	25.4						5.0 - 100.0
	10 - 90							

NUTRIENT		**Result**	**Low**	**Marginal**	**Sufficient**	**High**	**Excess**	**Sufficiency Range**
Colour (SM Code)	0 - 10	26.0						
Soil Texture (Hand bolus)	0 - 10	5.0						
Nitrate-N (Group) mg/kg	0 - 10	46						15 - 100
	10 - 90	3						15 - 100
Ammonium nitrogen (KCl) mg/kg	0 - 10	6						0 - 5
	10 - 90	2						0 - 5
Phosphorus (Colwell) mg/kg	0 - 10	170						20 - 30
Phosphorus (BSES) mg/kg	0 - 10	190						50 - 2,000
Phosphorus Buffer Index (Colwell) (PBIc)	0 - 10	300						15 - 840
Potassium (Amm-Acet.) cmol+/kg	0 - 10	1.90						0.40 - 0.50
Potassium % of CEC	0 - 10	7.5						1.0 - 10.0
Sulfate-S (KCl40) mg/kg	0 - 10	9.2						3.0 - 200.0
	10 - 90	2.1						3.0 - 200.0
Calcium (Amm-Acet) cmol+/kg	0 - 10	18.0						1.0 - 100.0
Calcium % of CEC	0 - 10	70.8						55.0 - 90.0
Magnesium (Amm-Acet.) cmol+/kg	0 - 10	5.5						0.5 - 20.0
Sodium (Amm-Acet.) cmol+/kg	0 - 10	0.03						0.00 - 1.00
Aluminium (KCl) cmol+/kg	0 - 10	0.1						0.0 - 0.5
Aluminium Saturation %	0 - 10	1.0						0.0 - 5.0
Copper (DTPA) mg/kg	0 - 10	3.00						0.30 - 5.00

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Zinc (DTPA) mg/kg	0 - 10	1.40	1.00 - 5.00
Manganese (DTPA) mg/kg	0 - 10	110.0	6.0 - 50.0
Iron (DTPA) mg/kg	0 - 10	90.0	
Boron (hot CaCl2) (mg/kg)	0 - 10	0.7	0.5 - 8.0

RECOMMENDATION: Lane

TIMING	PRODUCT	APPLICATION	RATE	QUANTITY
Post emergent	Urea S (kg/ha)	Broadcast	275 kg/ha	0.0t
At planting	Granulock Z (Incitec Pivot) (kg/ha)	Airseeder / seeding	60 kg/ha	0.0t
NUTRIENT RATES IN PRODUCT RECOMMENDATION				
NITROGEN (kg/ha)			116.6	
PHOSPHORUS (kg/ha)			13.1	
SULFUR (kg/ha)			16.2	
ZINC (kg/ha)			0.60	

PINNACLE AGRICULTURE

Grower Details:

Grower Name: FH Moses & Sons Pty Ltd
Paddock Name: Lane
Address: "Valais" Quirindi 2343
Phone:

Interpretation Date: Tuesday, 27 May 2025
Interpreter Name: Sam Gulliford
Interpreter Phone:

Sample Details:

Sample Date: Wednesday, 26 March 2025
Forecast Plant Date: 30-Apr-2025
Forecast Harvest Date: 15-Dec-2026
Test Date: Wednesday, 9 April 2025

Evaluation Parameters:

District: Quirindi NSW
Description: Lane
Crop: Canola
Soil Type: Black vertosol
Growth Stage:

Output Summary - DEMAND:

	YIELD		
	Low	**Expected**	**High**
Target Yield (t/ha):	2.4	3.0	3.6
Target Protein (%):	20	20.0	20
Target Oil (%):	43.0	43.0	43.0
Grain Removal:	77 kg/ha	96 kg/ha	115 kg/ha
Estimated Nitrogen Transfer Efficiency %:	45.0%	45.0%	45.0%
Estimated Crop N Demand (kg/ha):	171 kg/ha	213 kg/ha	256 kg/ha
Estimated N in Crop (kg/ha):			
N Required For Rest of The Crop (kg/ha)	171	213	256

Output Summary - SUPPLY:

Sample Profile Details:

Barcode	Depth From (cm)	Depth To (cm)	Bulk Density (t/m3)	Nitrate N (mg/kg)	Ammonium N (mg/kg)	Depth Weighting	Mineral N (kg/ha)	Organic Carbon (%)
070222295	0.00	10.00	1.20	46.00	5.50	1.00	58	3.79
070222294	10.00	90.00	1.30	3.40	1.90	1.00	35	

	Low	Expected	High
Soil Mineral N:	94 kg/ha	94 kg/ha	94 kg/ha
Rotation Credits:	0 kg/ha	0 kg/ha	0 kg/ha
Soil Mineralisation Credits:	168 kg/ha	168 kg/ha	168 kg/ha
Manure Credits:	0 kg/ha	0 kg/ha	0 kg/ha
Estimated Total Available N:	261 kg/ha	261 kg/ha	261 kg/ha
Recommended Nitrogen Rate:	**0 kg/ha**	**0 kg/ha**	**0 kg/ha**

Comments:

[Comments for depth 0.00 to 10.00]

[Comments for depth 10.00 to 90.00]

IMPORTANT NOTE: